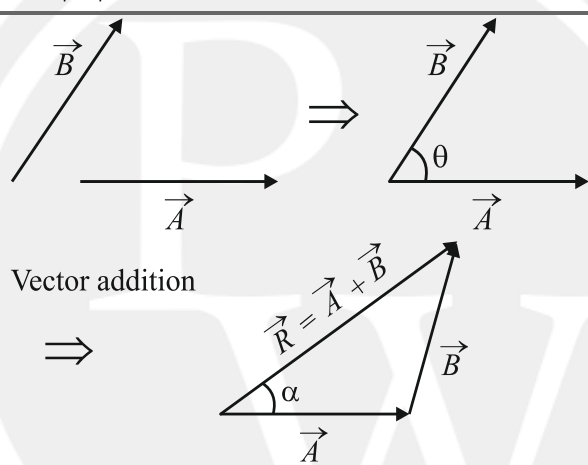
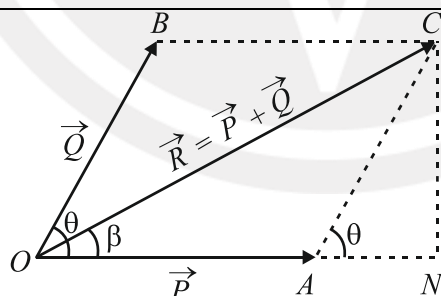


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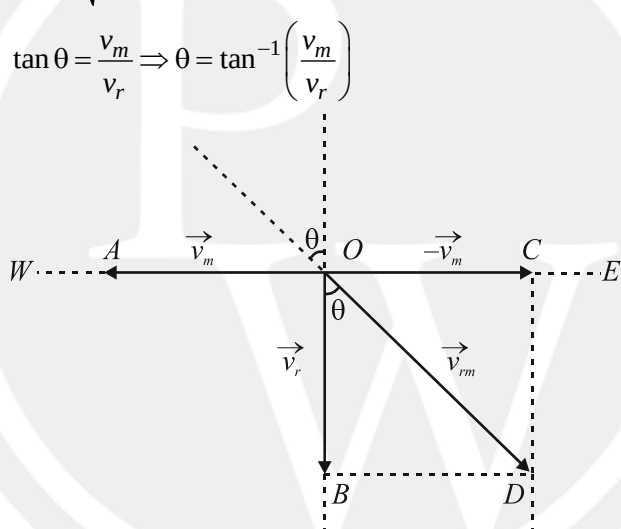
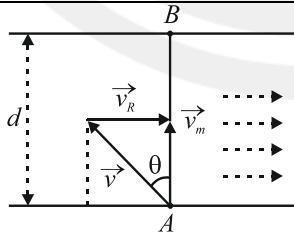
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Motion in Plane

Sr. No.	Concept	Formulas	Other information
1.	Unit vector	$\hat{A} = \frac{\vec{A}}{ \vec{A} }$, Modulus of vector $\vec{A} = \vec{A} = A$	
2.	Triangle law of vector addition	 Vector addition Resultant $\vec{R} = \vec{A} + \vec{B}$	$\vec{A}, \vec{B}$ = two vectors and angle between them is θ . α = angle between resultant vector $\vec{R}$ and $\vec{A}$ vector.
3.	Parallelogram law of vector addition	 $\vec{OA} + \vec{OB} = \vec{OC}$ or $\vec{P} + \vec{Q} = \vec{R}$ $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$ Where θ is the angle between $\vec{P}$ and $\vec{Q}$. If $\vec{R}$ makes angle β with $\vec{P}$, then $\tan \beta = \frac{Q \sin \theta}{P + Q \cos \theta}$	$\vec{P}, \vec{Q}$ = two vectors are represented by two adjacent sides of a parallelogram which are directed away from their common point. R = magnitude of the resultant $\vec{P}$ and $\vec{Q}$ vector.

Sr. No.	Concept	Formulas	Other information
4.	Vector subtraction	$ \vec{A} - \vec{B} = \sqrt{A^2 + B^2 + 2AB\cos(\pi - \theta)}$ $= \sqrt{A^2 + B^2 - 2AB\cos\theta}$ $\& \tan\alpha = \frac{B\sin\theta}{A - B\cos\theta}$	θ = angle between $\vec{A}$ and $\vec{B}$. α = angle between resultant vector and $\vec{A}$ vector.
5.	Properties of vector addition	Vector addition is commutative; $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ Vector addition is associative; $(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$	
6.	Rectangular components of a 3D vectors and direction cosines	$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$, $ \vec{A} = \sqrt{A_x^2 + A_y^2 + A_z^2}$ $\cos\alpha = \frac{A_x}{A} = \frac{A_x}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = l$ $\cos\beta = \frac{A_y}{A} = \frac{A_y}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = m$ $\cos\gamma = \frac{A_z}{A} = \frac{A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = n$ $l^2 + m^2 + n^2$ $= \cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{A_x^2 + A_y^2 + A_z^2}{(\sqrt{A_x^2 + A_y^2 + A_z^2})^2} = 1$ or $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.	l, m, n = direction cosines of vector $\vec{A}$. α, β, γ = angles made with x, y and z axis respectively. A_x, A_y, A_z = three rectangular components of $\vec{A}$.
7.	Scalar product	$\vec{A} \cdot \vec{B} = AB\cos\theta$ Angle between two vectors $\theta = \cos^{-1}\left(\frac{\vec{A} \cdot \vec{B}}{AB}\right)$ $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$, $\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = 0$	$\hat{i}, \hat{j}, \hat{k}$ = unit vectors along x, y and z axis.
8.	Cross product (Vector product)	(i) $\vec{A} \times \vec{B} = AB\sin\theta\hat{n}$ $\vec{A} \times \vec{B} = \hat{i}(A_yB_z - A_zB_y) - \hat{j}(A_xB_z - B_xA_z) + \hat{k}(A_xB_y - B_xA_y)$ $\hat{i} \times \hat{i} = \vec{0}$, $\hat{j} \times \hat{j} = \vec{0}$, $\hat{k} \times \hat{k} = \vec{0}$, $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (ii) If two vectors are parallel or antiparallel then $\vec{A} \times \vec{B} = \vec{0}$ (iii) $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$ (iv) Unit vector perpendicular to $\vec{A}$ and $\vec{B}$, $\vec{C} = \frac{\vec{A} \times \vec{B}}{ \vec{A} \times \vec{B} }$ (v) $ \vec{A} \times \vec{B} $ = Area of parallelogram whose two sides are represented by two vectors $\vec{A}$ and $\vec{B}$.	θ = angle between two vectors $\vec{A}$ and $\vec{B}$

Sr. No.	Concept	Formulas	Other information
9.	Projectile fired at an angle θ with the horizontal. (Oblique projectile motion)	(i) Components of initial velocity: $u_x = u \cos \theta$, $u_y = u \sin \theta$ (ii) Components of acceleration at any instant: $a_x = 0$, $a_y = -g$ (iii) Position after time t: $x = (u \cos \theta)t$, $y = (u \sin \theta)t - \frac{1}{2}gt^2$ (iv) Equation of trajectory: $y = x \tan \theta - \frac{g}{2u^2 \cos^2 \theta} \cdot x^2$ (v) Maximum height, $H = \frac{u^2 \sin^2 \theta}{2g} = \frac{u_y^2}{2g}$ (vi) Time of flight, $T = \frac{2u \sin \theta}{g} = \frac{2u_y}{g}$ (vii) Horizontal range, $R = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x u_y}{g}$ (viii) Maximum horizontal range is attained at $\theta = 45^\circ$ and its value is $R_{\max} = \frac{u^2}{g}$ (ix) Velocity after time t: $v_x = u \cos \theta$, $v_y = u \sin \theta - gt$ $\therefore v = \sqrt{v_x^2 + v_y^2}$ and $\tan \beta = \frac{v_y}{v_x}$	u = projected with velocity at an angle θ. u_x = horizontal component of initial velocity u_y = vertical component of initial velocity a_x = horizontal component of acceleration a_y = vertical component of acceleration β = the angle made by velocity vector with horizontal at that instant.
10.	Horizontal projection from a height h	Position after time t: $x = ut$, $y = \frac{1}{2}gt^2$ Path of projectile. $y = \left(\frac{g}{2u^2} \right) x^2$ Velocity of the object at any instant $v = \sqrt{v_x^2 + v_y^2} = \sqrt{u^2 + g^2 t^2}$ Time of flight $T = \sqrt{\frac{2h}{g}}$ Horizontal range $R = uT = u \sqrt{\frac{2h}{g}}$ Angle made by velocity at any instant with horizontal $\theta = \tan^{-1} \left(\frac{gt}{u} \right)$	u = initial horizontal velocity of projectile h = height of projection tower g = gravitational acceleration

Sr. No.	Concept	Formulas	Other information
11.	Relation between H, R and T in projectile motion and trajectory equation	$H = \frac{u_y^2}{2g} = \frac{u^2 \sin^2 \theta}{2g} = \frac{1}{8} g T^2, \quad \frac{H}{R} = \frac{1}{4} \tan \theta$ Horizontal range is same for θ and $(90^\circ - \theta)$ and maximum for $\theta = 45^\circ$. Equation of trajectory $y = x \tan \theta \left(1 - \frac{x}{R} \right)$	H = maximum height attained by the projectile R = horizontal range, T = time of flight, θ = angle of projection, x = position of the particle at any time t .
12.	Relative velocity when two bodies are moving at an angle θ.	$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$ $v_{AB} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos \theta}$ $\tan \beta = \frac{v_B \sin \theta}{v_A - v_B \cos \theta}$	A, B = bodies moving with velocity $\vec{v}_A$ and $\vec{v}_B$ inclined at an angle θ . v_{AB} = magnitude of the relative velocity of A w.r.t. B is given by.
13.	Rain-man problem	$v_{rm} = \sqrt{v_r^2 + v_m^2 - 2v_r v_m \cos 90^\circ} = \sqrt{v_r^2 + v_m^2}$ $\tan \theta = \frac{v_m}{v_r} \Rightarrow \theta = \tan^{-1} \left(\frac{v_m}{v_r} \right)$ 	v_{rm} = velocity of rain with respect to man. v_r = velocity of rain with respect to ground. v_m = velocity of man with respect to ground.
14.	Boat river problems	 For shortest path: $v \sin \theta = v_R \Rightarrow \sin \theta = \frac{v_R}{v}$ Time of crossing $t = \left(\frac{d}{v \cos \theta} \right)$	v = velocity of the swimmer. v_R = velocity of the river. θ = angle made by the man with river flow. $t_{\min}$ = minimum time to cross the river. v_m = resultant velocity.
15.	Angular displacement (θ)	$\theta (\text{rad}) = \frac{\text{Arc length}}{\text{Radius}} = \frac{s}{r}$	

Sr. No.	Concept	Formulas	Other information
16.	Angular velocity (ω) and Time period	$\omega = \frac{\theta}{t} \text{ or } \omega = \frac{\theta_2 - \theta_1}{t_2 - t_1}, \omega = \frac{2\pi}{T} = 2\pi \nu$ $\nu \text{ is the frequency of revolution. } \left(\nu = \frac{1}{T} \right)$	ω = angular velocity. T = time taken for one complete revolution (time period)
17.	Some important relations of circular motion	$v = r\omega$ i.e. Linear velocity = Radius $\times$ Angular velocity. $\alpha = \frac{\omega_2 - \omega_1}{t_2 - t_1}, \text{ Also, } a = r\alpha$ i.e. Linear acceleration = Radius $\times$ Angular acceleration $a = \frac{v^2}{r} = r\omega^2$	$(\omega_2 - \omega_1)$ = change in angular velocity a = centripetal acceleration r = radius of the circular path α = angular acceleration

